

Clinic no-shows: fix the reminder that silently stops working before trying to predict who won't come

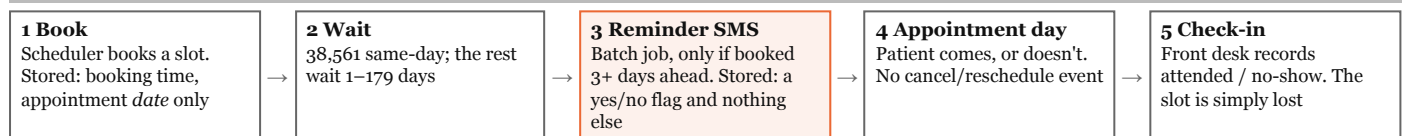
One in five booked appointments in Vitória's public clinics goes unused. The client wants an AI model to predict no-shows. The data says something else is broken first. On **7 of 27 appointment days**, not one patient booked in advance got an SMS reminder, nobody noticed, and on those days no-shows rose by **+3.9 points** while the control group didn't move. A dependable, monitored reminder step comes before any predictor.

The situation. The request is my assumption: a municipal health secretariat saying “20.2% of patients don't turn up. Build us a model that predicts no-shows so we can overbook.” The data is real: 110,527 appointments from the public health system in Vitória, Brazil, covering 29 Apr–8 Jun 2016 (Kaggle “Medical Appointment No Shows”, CC BY-NC-SA). After cleaning that is 110,516 appointments, 62,296 patients, 81 clinic neighbourhoods and about **826 empty slots per working day**. I have not interviewed anyone. Everything below comes from the data, and where I'm guessing I say so.

1 Who is involved

Who	Role	What they want / what limits them	What I would ask first
Municipal health secretariat	Owens the outcome, pays	More patients seen for the same doctors; no budget for extra staff	What would make you stop this after a month?
Clinic front desk / schedulers	Operator	Fill the day, fewer idle doctors; they have no time to phone every patient	Walk me through the last day half the list didn't come.
IT / SMS vendor	Owens the reminder job	Keep the nightly batch running; SMS costs money per message	Where is the log of what was sent and delivered on 16 May?
Doctors	Affected	Lose consultation time to empty slots; dislike overbooking	What happens today when three patients don't show?
Patients	Affected, user	Long waits for a slot; many on welfare, no car, work shifts	How did you know about your appointment? Did you get a message?

2 What happens today, and where it breaks



Where it breaks (step 3): reminders stop without anyone noticing, and 18% of eligible bookings go unreminded even on normal days. **Missing events:** SMS sent/delivered time, the patient's phone number on file, appointment time of day, cancellations. Without these, nobody can see step 3 failing.

3 Facts, assumptions, unknowns

Fact (measured in the pipeline)	Assumption (believed, not proven)	Unknown (need a person or a log)
- Same-day bookings no-show 4.6%; 3+ day bookings 29.7%. - No booking made under 3 days ahead ever got an SMS. - SMS coverage of eligible bookings: 59% overall, 82% on normal days, 0% on 9 May, 13 May and every day of 16–20 May. 24,536 eligible bookings got no SMS: 16,581 on outage days, 7,955 on normal days. - Rain made no difference: 20.3% dry, 20.4% light, 18.8% heavy (2 days).	- The request and stakeholders above (the data is real, the client is not). - The 3-day threshold is policy. I inferred it from the data; nobody told me. “Neighbourhood” is where the clinic is, as the dataset notes say. - The outage days have nothing to do with which patients booked, so they work as a natural experiment. 2016 behaviour is similar enough to today's to act on.	- Were SMS really not sent on outage days, or only not recorded? The evidence points to not sent (next page). - Why do 18% of eligible bookings miss out on normal days? Missing phone numbers? - Why no data on 23 May and 27 May? The holiday API only explains Corpus Christi. - Are the 16,200 same-patient, same-day rows double bookings or two specialties?

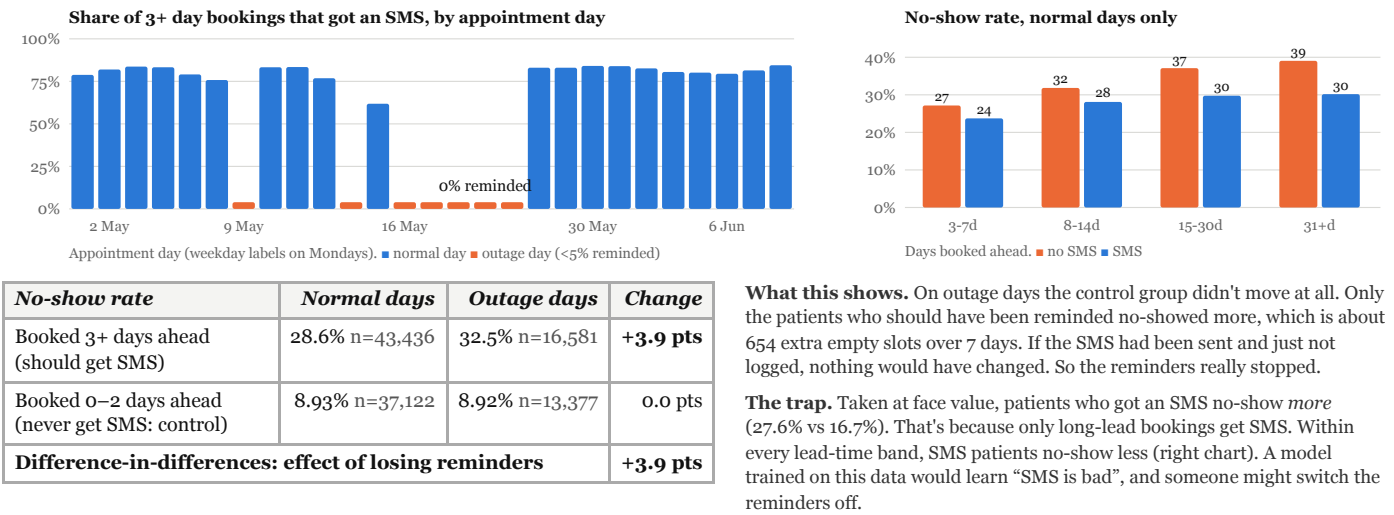
4 The problem, and how success is measured

Problem statement. Patients who book 3+ days ahead in Vitória's public clinics miss 29.7% of appointments. The SMS reminder that should prevent this reaches only 59% of them, and it silently failed on 7 of 27 days because nothing monitors it. **Goal:** within 8 weeks, get daily reminder coverage to at least 95% on every day, with a same-day alert when it drops, and bring the advance-booking no-show rate below 28%. No change to booking rules, clinic capacity or staffing.

Type	Metric (all computed by the pipeline's SQL)	Today	Target, 8 weeks
Project KPI	No-show rate, appointments booked 3+ days ahead (lagging)	29.7% of 60,017	< 28%
Supporting	Reminder coverage of eligible bookings, checked daily (leading)	59%	≥ 95% every day
Supporting	Outage days (coverage < 5%), and hours until someone notices	7 of 27; never	0; same day
Supporting	Reminders with a send time and delivery status on record	0% (flag only)	100%
Supporting	Weekdays missing from the extract with no explanation	2	0
Guardrail	Same-day no-show rate (never reminded, so it should not move)	4.6%	no change

Why <28% and not lower: removing the outages alone is worth about 1.1 points (654 no-shows on 16,581 bookings), and closing the normal-day gap is worth roughly 0.6 more. Reminders will not fix most no-shows. They are the part that is broken, and the part everything else would depend on.

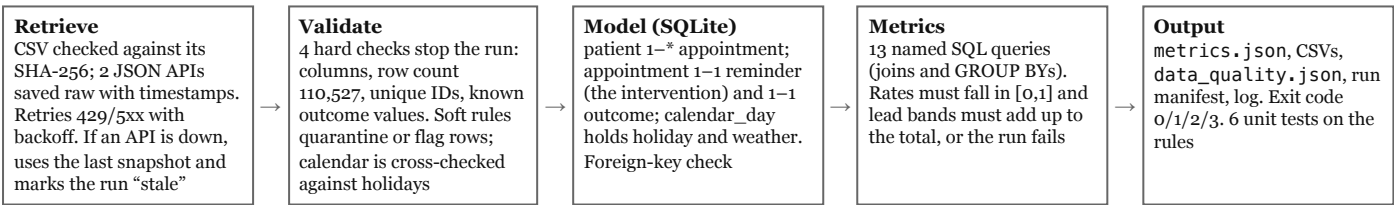
5 The evidence



6 Where the truth lives, and how I got it

Information need	Source used (type)	System of record / owner	Trust and gaps
Booking and appointment date	Appointments extract (CSV, real, 10.7 MB, SHA-256 checked)	Scheduling system / secretariat IT	Good. The appointment has a date but no time, so lead time is in whole days
Was a reminder sent?	SMS_received flag in the same CSV	SMS gateway log / vendor (not available)	Weak: a copied flag with no time or delivery status. This is the key missing event
Did the patient attend?	No-show column in the CSV	Front-desk check-in	Good, but the label reads backwards: "No" means the patient came
Was the clinic open?	Nager.Date public holidays (JSON API)	Municipal calendar / secretariat	National holidays only. It explains 1 of the 3 missing weekdays (23 May, 26 May, 27 May)
Did weather matter?	Open-Meteo archive, Vitória (JSON API)	Context only, not a system of record	Gridded reanalysis rather than a weather station; fine for testing a claim

7 The pipeline: retrieve → validate → model → metrics



Found and handled: 11 rows quarantined (5 booked after their appointment; ages –1 and 115); the outcome label inverted into no_show (1 = absent); misspelled columns renamed; Handcap kept as a 0–4 count, not a yes/no; same-day bookings made after midnight fixed to lead 0, not –1; 16,200 same-patient, same-day rows flagged but kept; 39 Saturday appointments kept. Code, raw data and run logs are in the accompanying repo (noshow-pipeline).

8 What I would do first, and what I would leave alone

- MVP, 8 WEEKS, NO AI**

 - A daily coverage check on the reminder job: eligible bookings vs SMS sent, alert the same day it falls below 95%.
 - Get the SMS vendor's send/delivery log into the pipeline, so the flag gets replaced by the real event.
 - List the eligible patients with no phone number on file and fix those at the front desk.
 - Check the batch cut-off: bookings made exactly 3 days ahead get 53% coverage on normal days, against about 85% at 4–5 days.
- DELIBERATELY OUT OF SCOPE**

 - A no-show prediction model or overbooking. The data would teach it that SMS is harmful, and an outage looks just like patient behaviour.
 - Shorter wait times, even though long waits are the bigger driver (4.6% vs 29.7%). That needs capacity data I don't have.
 - Targeting the 3,222 repeat no-show patients (5.2% of patients, 35% of no-shows) until reminders work.
 - Replacing the scheduling system.

9 What must be proven before going further, and why this problem first

Question	How I'd check	Carry on if
Were SMS actually not sent on 9, 13 and 16–20 May?	Vendor send logs for those dates; ask IT what changed that week	The logs confirm the gap
Does the effect hold going forward?	Switch the monitoring on in half the clinics first and compare the KPI	≥ 1 point lower within 4 weeks
Why do 18% miss out on normal days?	Join to the patient registry: phone number present? valid?	A fixable cause covers most of them

Why this is the right problem to solve first. It is the one step in the workflow that is *measurably broken*, and a natural experiment in the data shows that fixing it helps. It is cheap (a count and an alert), quick to try, and easy to undo. And it comes before anything smarter: a prediction model would be trained on labels the broken reminder step has already corrupted, and any action that model triggered, like an extra message or a call, would go out through the same unmonitored step. Fix the thing that silently fails, then see what is left to predict.